

10.023 Designing Energy System (DES)
1D Report Worksheet (2025)

Group: 3.2
Due Date: Week 13 Friday (15 Aug) 6 pm

Grading

This 1D is worth a total of 20% of the total marks allocated for DES based on: (1) 1D worksheet 1 (3%); (2) 1D worksheet 2 (3 %); (3) product exhibit (6%); and **(4) worksheet consisting of data collection and analysis (8%)**.

Maximum 50 words Problem Statement (Include background, clear problem description, evidence, impact, and desired outcome to define and explain the issue effectively)

Due to increased death rates in Singapore Army, SAF uses Wet Bulb Globe Thermometers (battery life ~1 week) for heat injury prevention. Human battery management leads to forgotten replacements, with strict enforcement on training halts. A reliable, automated power solution is needed to ensure continuous monitoring and avoid operational disruptions.

In no more than 50 words, describe your solution clearly. Explain how it addresses the problem.

With the enhancement with a solar panel to the Wet Bulb Globe Thermometer (WBGT), coupled with its natural outdoor use, the charging process can be made automatic during usage, removing the chance of human error. This ensures training time can be maximized with the higher availability of WBGT monitor.

Explain each modification clearly (reader should be able to replicate from this instruction).

*** Refer to schematic on page 6 and 7 for visualization.**

Electrical:

Change of Battery (Two AAA Batteries – One ML2032)

- AAA Batteries are not rechargeable, and capacity can be reduced with the addition of a solar panel. After calculations, One ML2032 (3V ~ 3.2V), 65 mAh would suffice to power the device.

Solar Panel Installation

- WBGT Battery Pin to Solar Charge Controller
Solar Charge Controller load pins should supply power to the device instead of the battery directly. Hence, there is a need to disconnect the WBGT Board Battery Pins from the existing battery holder and connect the board to the load pins of the charge controller's B+ and B- respectively.
- Solar Panel to Solar Charge Controller
Solar Panel load should supply power to the Controller. Hence, there is a need to connect the Solar Panel's + and - to the Controller's T+ and T- respectively.
- New Battery to Solar Charge Controller
New Battery should be able to charge through and power load through the Controller. Hence, there is a need to connect the new batteries' + and - to the Controller's B+ and B- respectively.

Mechanical:

Solar Charge Controller Fitting

- The additional solar charge controller board is connected to all other components, including the existing WBGT circuit board, the new battery, and new solar panel. Thus, it is placed near the center of the newly designed and 3D printed casing and held in place by extrusions on its base.

Battery and Holder Fitting:

- The battery is placed in the battery holder, where it is held in place by extrusions on the 3D printed housing.

Solar Panel Fitting:

- An adjustable arm allows the solar panel to be unfolded when the battery needs charging. It can also be folded back onto the main housing, keeping the device portable.

Device Stand:

- The whole device is set up on a stand, where it is threaded onto the 3D printed case and raised from the ground where it can achieve accurate readings from its sensors.

In this one page, provide details of data collection process and why the data is important.

	<u>Power Consumption</u> (I, V)	<u>Context</u> (hours/day & day/week)	<u>Solar Charge Controller States</u> (V and I across for all component)	<u>Irradiance</u> (Voc, Isc, V and I)										
Steps Taken:	1. Under simulated use case (In the hot sun) we placed our device WBGT under it. 2. Measure the voltage of the device by connecting a DMM in parallel across the battery. 3. Break the connection between battery and device circuit board and connect the DMM in series to measure current. *Shown in Figure 1	1. Interview Officers (Training planners) on their use case of WBGT 2. Find out the average duration in which WBGT measurement is taken on a daily and weekly basis 3. Validate existing battery management problem.	Under 4 different conditions: <table><tr><td>Halogen Light</td><td>On Board Switch</td></tr><tr><td>ON</td><td>ON</td></tr><tr><td>ON</td><td>OFF</td></tr><tr><td>OFF</td><td>ON</td></tr><tr><td>OFF</td><td>OFF</td></tr></table> 1. Measure voltage across solar panel, battery and load (WBGT) 2. Measure current in series between solar panel, battery, load and charge controller. *Refer to Figure 2 for result. <u>Conclusion:</u> - On board switch when ON facilitates the powering of load by the battery only. - Charging occurs whenever the solar panel under light produce voltage of 3V.	Halogen Light	On Board Switch	ON	ON	ON	OFF	OFF	ON	OFF	OFF	1. Under halogen lamp we varied the height away from from the solar panel to achieve intervals of irradiance 400, 600, 800, 1000 W/m^2. 2. At each interval, measure Voc and Isc by shorting the solar panel with the DMM and measure its voltage and current. 3. Connect Solar Panel to charge controller. 4. At each interval measure voltage across battery connecting DMM in parallel. 5. At each interval measure current from battery connecting DMM in series. *Refer to Figure 3 for result.
Halogen Light	On Board Switch													
ON	ON													
ON	OFF													
OFF	ON													
OFF	OFF													
Purpose	To find out how much power (calculated using voltage and current output) our WBGT consumes so that we can size suitable battery and solar panel.	To find out the daily load (max current draw x duration) consumed so that we can size suitable battery and solar panel.	Due to the uncommon and minimal specification found online of the solar charge controller. Further testing was needed to verify its use case.	To measure the solar performance of the added solar powered system, and calculate charging feasibility (Charging rate > Load Power Consumption)										

In this **one** page, detail the energy and power analysis of the device before and after modification. Explain the results.

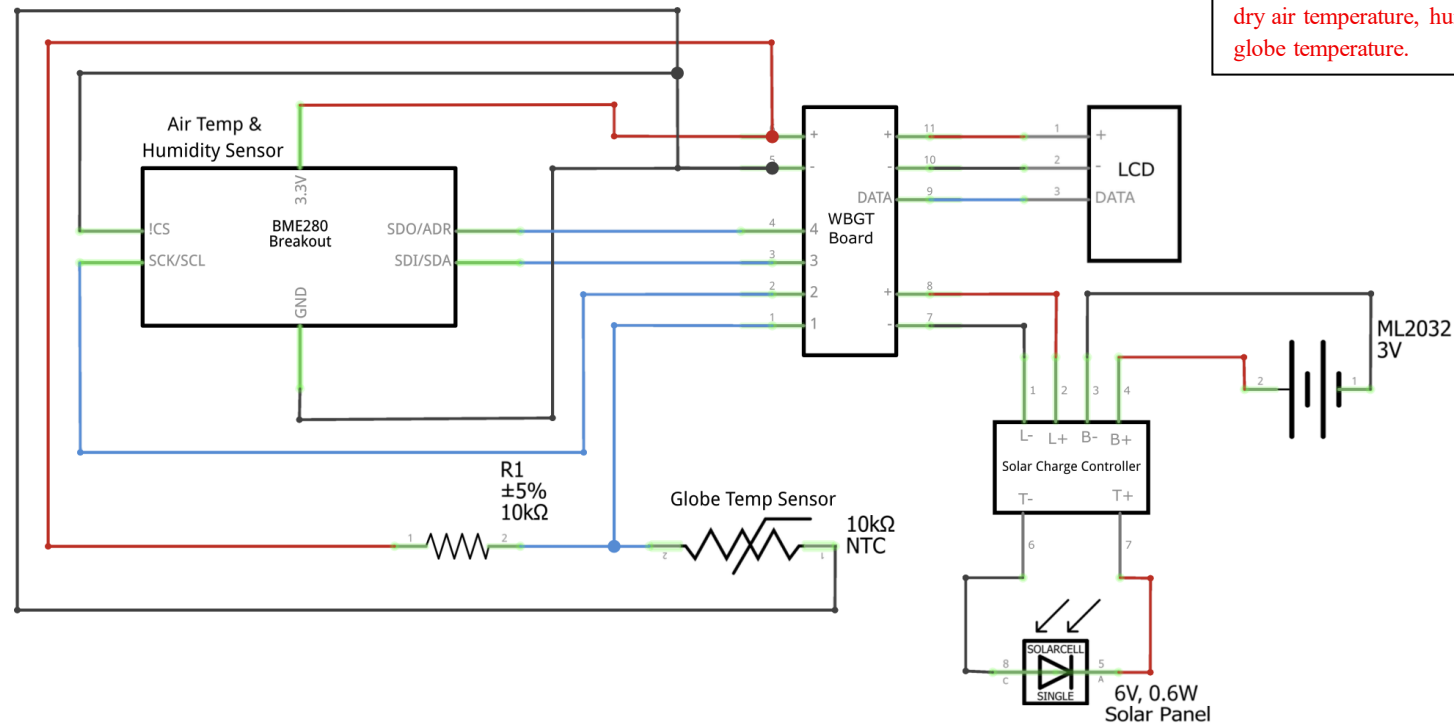
Before	After																									
Power Consumption Device Voltage: 2.553V (Recorded) , 3V (Theoretical) Device Current Draw: 290μA *As seen in Figure 1	Power Consumption Device Voltage: 3.31V Device Current Draw: 150μA (Reduced due to not triggering buzzer) *As seen in Figure 6																									
Context Weekly Usage: ~ 5 days a week Daily Usage: 6 hrs/day (breaks during high WBGT Reading for safety) Total Daily Load: 290μA x 6 h = 1.74mAh	Solar Panel Performance <table><tr><th>Irradiance (W/m^2)</th><th>Voc (V)</th><th>Isc (mA)</th><th>Battery Voltage (V)</th><th>Battery Current (mA)</th></tr><tr><td>400</td><td>4.915</td><td>22.59</td><td>4.591</td><td>45.91</td></tr><tr><td>600</td><td>4.9</td><td>34.94</td><td>4.75</td><td>47.5</td></tr><tr><td>800</td><td>4.936</td><td>45.35</td><td>4.862</td><td>48.62</td></tr><tr><td>1000</td><td>4.915</td><td>55.99</td><td>5.042</td><td>50.42</td></tr></table>	Irradiance (W/m^2)	Voc (V)	Isc (mA)	Battery Voltage (V)	Battery Current (mA)	400	4.915	22.59	4.591	45.91	600	4.9	34.94	4.75	47.5	800	4.936	45.35	4.862	48.62	1000	4.915	55.99	5.042	50.42
Irradiance (W/m^2)	Voc (V)	Isc (mA)	Battery Voltage (V)	Battery Current (mA)																						
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800	4.936	45.35	4.862	48.62																						
1000	4.915	55.99	5.042	50.42																						
Battery Sizing *(Refer to full calculation at Figure 5) Unadjusted battery capacity Days of Autonomy x Load: 0.0087 Ah Capacity adjusted for MDOD: 0.0435Ah (never below 20% MDOD) Capacity adjusted for MDDOD: 0.0174Ah (never below 10% MDDOD) Capacity adjusted for EOL: 0.010875Ah (80% EOL remaining) Capacity adjusted for design margin (1.1): 0.04785Ah However, the smallest commercially available unit that exceeded this requirement is a 65 mAh ML2032 3V cell the practical choice. *Screenshot of Datasheet in Figure 8	*As seen in Figure 3 <u>Voc</u> : Expected small increase over increasing irradiance. Not achieved possibly due to multimeter accuracy. <u>Isc</u> : Result is as expected with almost linear increase with increasing irradiance. <u>Battery Voltage and Current</u> : Increase linearly with increasing radiance as expected. [Disclaimer] Values which was supposed to be used to measure charging capabilities, although possible, is not recommended due to solar charge controller’s misinformation of its voltage regulating feature. It is unable to down regulate solar input voltage from > 4.5 V to 3.2 V for battery. *Images of false specifications in Figure 9																									
Solar Panel Sizing *(Refer to full calculation at Figure 5) Chosen PV module: Alilin Solar Panel 6V, 0.6W (Website) Daily device energy draw: 0.00172Ah Lowest Average Peak Sun Hour in 2024: 3.87 hrs *Figure 4 Daily Ah of panel, adjusted after losses and A:L: 0.27132Ah No. of panels required: 0.008337019 ≈ 1 Dividing the adjusted requirement by the panel’s daily output shows that a single panel comfortably exceeds the needed capacity. Lower voltage panel like 4.5V ones were not selected due to solar charge controller’s specifications for input panel voltage be > 1.5x of battery voltage. *Images of false specifications in Figure 9	Charging Feasibility [Disclaimer] Assuming voltage regulation properly reduce voltage. Realistic calculation included. During the hottest time of day where training halts for safety reason, 12-2pm, according to suncalc website , the irradiance value can will be approximately 1000 W/m^2. Capacity charged in 2 hours = 50.42 mA x 2 h = 100.84 mAh Total Daily Load = 1.74 mAh (Capacity Charged > Total Daily Load) MDOD = 65 mAh * 20% = 13 mAh (Capacity Charged > MDOD) * Charging rates for different irradiance can be seen in Figure 7 Capacity charged in 2 hours greatly surpass the total daily load and its rated MDOD in mAh. Showing its feasibility in charging the battery during stipulated training pause.																									

In no more than 50 words, discuss the impact on functionality and usability of the proposed solution.

Addition of solar panel, the monitor can be charged outdoors during breaks (where the weather is too hot for training but bright enough to charge). Soldiers do not have to manage battery replacement. Device designed to have limited addition in size, can be fitted into a uniform pocket for portability.

Detailed schematic of the working prototype (mechanical and/or electrical).

Electronics Schematics

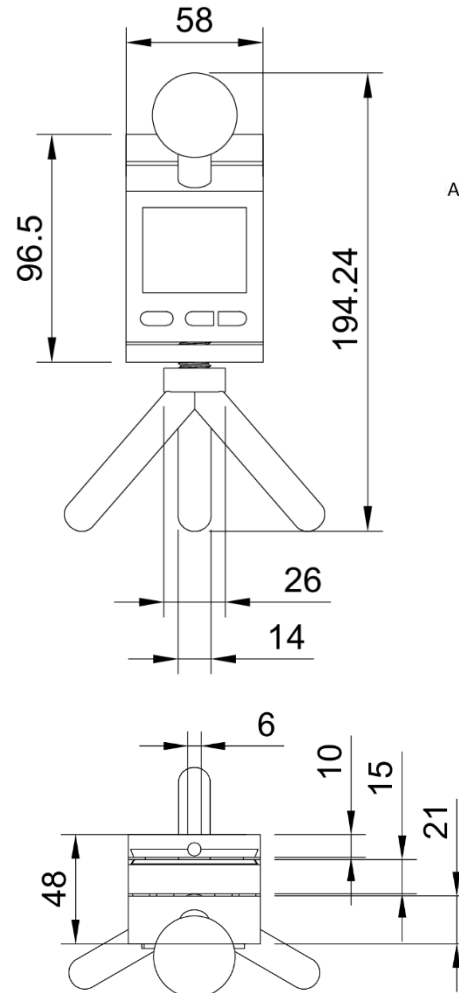
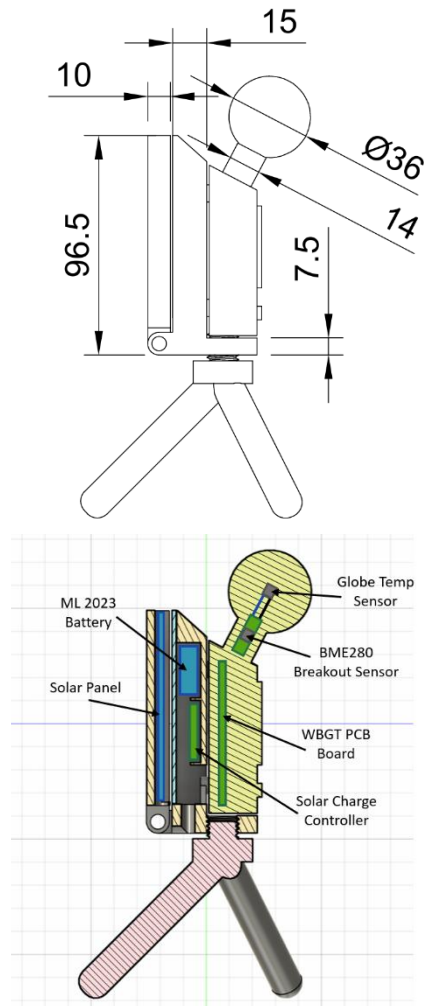


* LCD Connection is simplified to two power line and one data line.
* Wet Bulb Globe Thermometer value is mathematically computed based on dry air temperature, humidity and black globe temperature.

* LCD connection is simplified

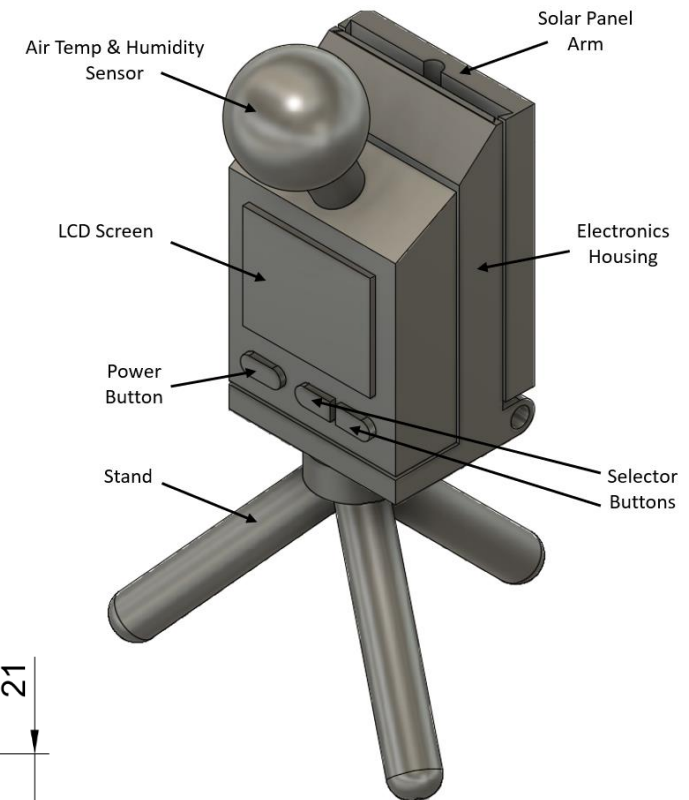
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Mechanical Schematic



Design Consideration:

- Portable
- Ease of Use
- Preserving Functionality



Photos of working prototype

Working in Outdoors	Charging in Outdoors (Front)	Charging in Outdoors (Back)
		
	Charging in Indoor (LED Light Up)	
		

References:

1. Lydia, L. (2021, January 27). *Dave Lee's death from heat stroke 'truly a sad loss': Coroner on SAF case*. Channel News Asia. <https://www.channelnewsasia.com/singapore/dave-lee-death-heat-stroke-coroner-truly-a-sad-loss-saf-435246>
2. Shopee. (n.d.). *WBGTHI Heat Index Checker Stress Meter Air Globe Temperature Humidity Tester*. Shopee. https://shopee.sg/casytw-WBGT-HI-Heat-Index-Checker-Stress-Meter-Air-Globe-Temperature-Humidity-Tester-i.642826180.41557245463?sp_atk=a9d224a3-aa1b-45ba-be27-94be30462015&xptdk=a9d224a3-aa1b-45ba-be27-94be30462015
3. Shopee. (n.d.). *Alilin Solar Panel 6V Polysilicon Module Easy to Carry for DIY Educational Use*. Shopee. https://shopee.sg/Alilin-Solar-Panel-6V-Polysilicon-Module-Easy-to-Carry-for-DIY-Educational-Use-i.1333197929.40603532678?sp_atk=75bd430d-e29f-48d6-b282-137581199de1&xptdk=75bd430d-e29f-48d6-b282-137581199de1
4. Shopee. (n.d.). *3.2V-15V Solar Light Circuit Control Board Solar Garden Light Controller PCBA Board*. Shopee. https://shopee.sg/3.2V-15V-Solar-Light-Circuit-Control-Board-Solar-Garden-Light-Controller-Light-Control-Board-PCBA-Board-i.1107145475.28837002408?sp_atk=68189926-7e8f-4b9e-b5ff-223d7400d5ea&xptdk=68189926-7e8f-4b9e-b5ff-223d7400d5ea
5. NASA POWER. (n.d.). *NASA POWER data access viewer*. National Aeronautics and Space Administration. <https://power.larc.nasa.gov/data-access-viewer/>
6. SunCalc. (n.d.). *Sun position, sunlight phases, moon position, and lunar phase calculator*. SunCalc. <https://www.suncalc.org/#/1.348,103.9463,14/2025.08.13/12:00/1/3>

Appendix:

Figure 1: Data collection process (Voltage and Current under the sun)

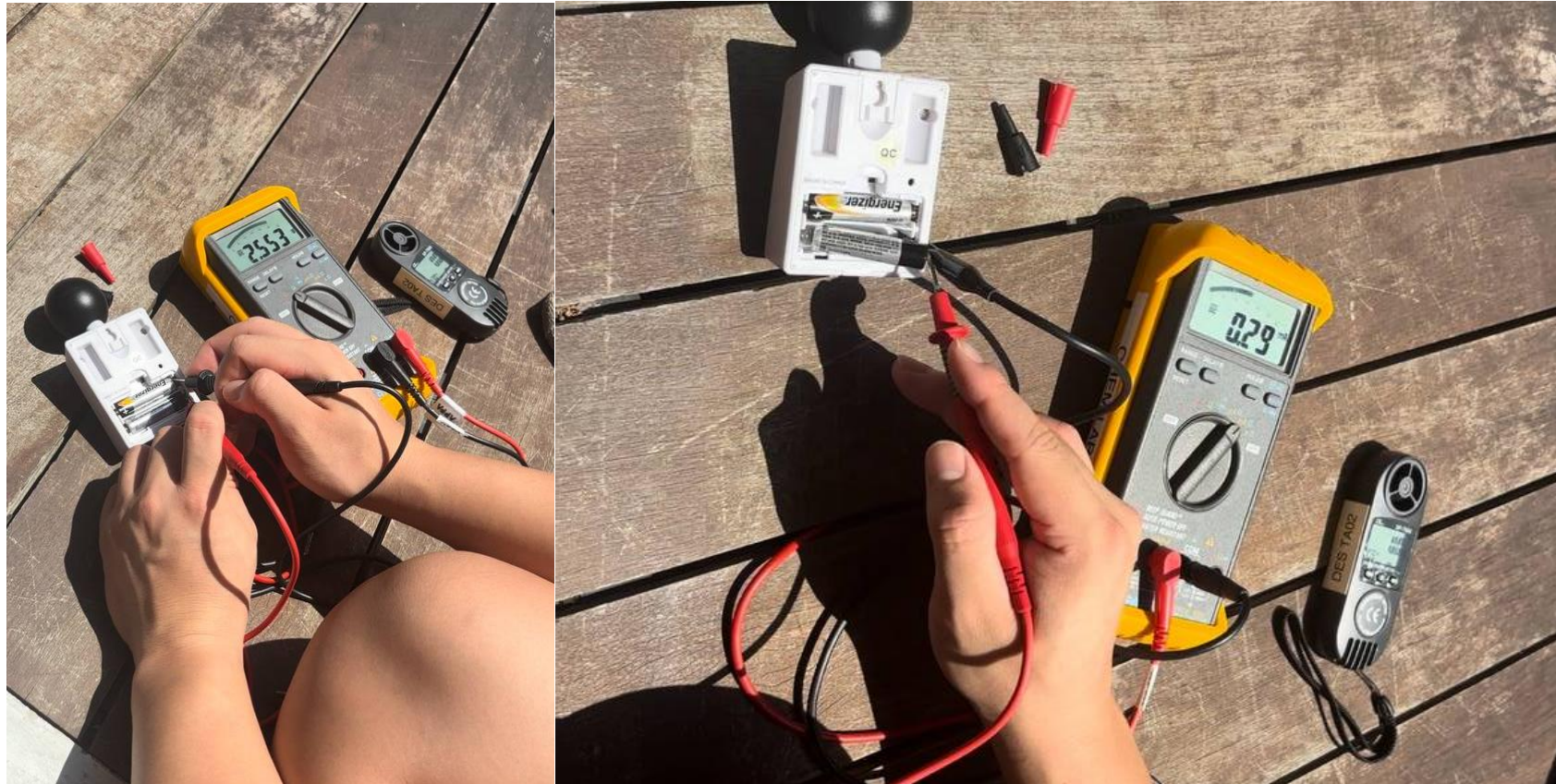


Figure 2: V and I of Different Components in Varying Conditions

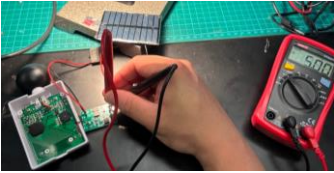
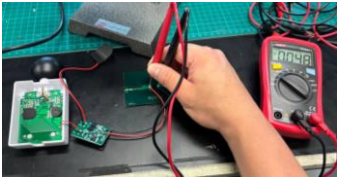
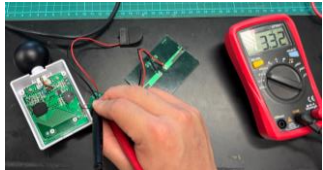
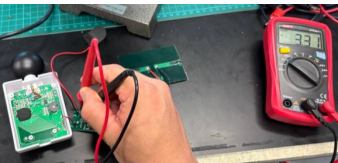
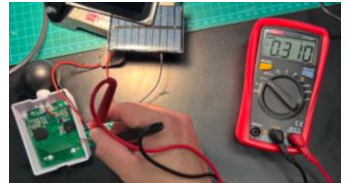
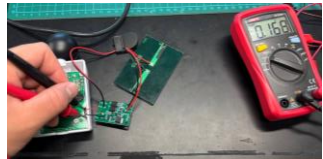
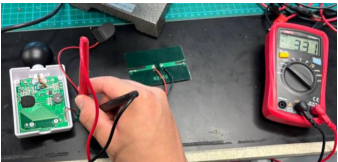
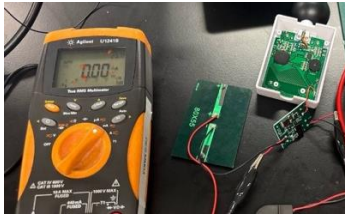
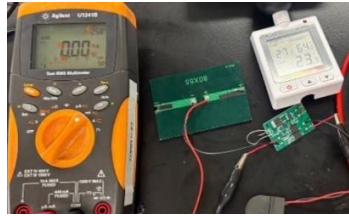
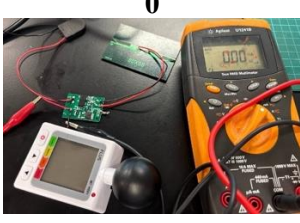
	Light (SW On)	Light (SW Off)	No Light (SW Off)	No Light (SW On)
Solar (V)	5.04 	5 	0.113 	0.048 
Battery (V)	4.76 	4.78 	3.32 	3.31 
Load (V)	0.31 	0.195 	0.168 	3.31 

Figure 2 Cont'd: V and I of Different Components in Varying Conditions

	Light (SW On)	Light (SW Off)	No Light (SW Off)	No Light (SW On)
Solar (I)	30.75mA 	34.42mA 	0 	0 
Battery (I)	-29.45mA 	-31.11mA 	0.02mA 	0.15mA 
Load (I)	0 	0 	0 	0 
Battery State	Charging	Charging	Not Charging	Discharging

*Current drawn by load shows zero throughout due to relatively high resistance in DMM compared to the device itself.

Figure 3: Isc, Voc, I and V at Varying Irradiances

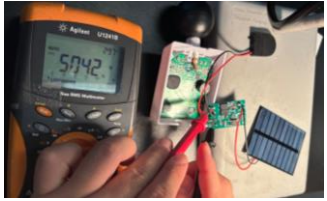
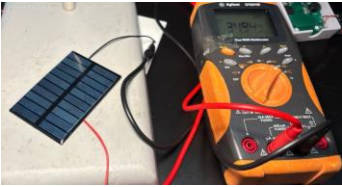
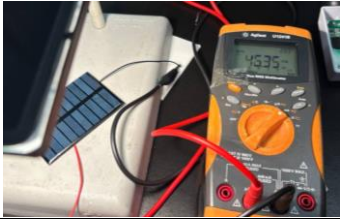
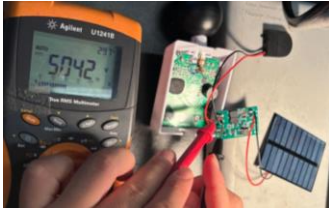
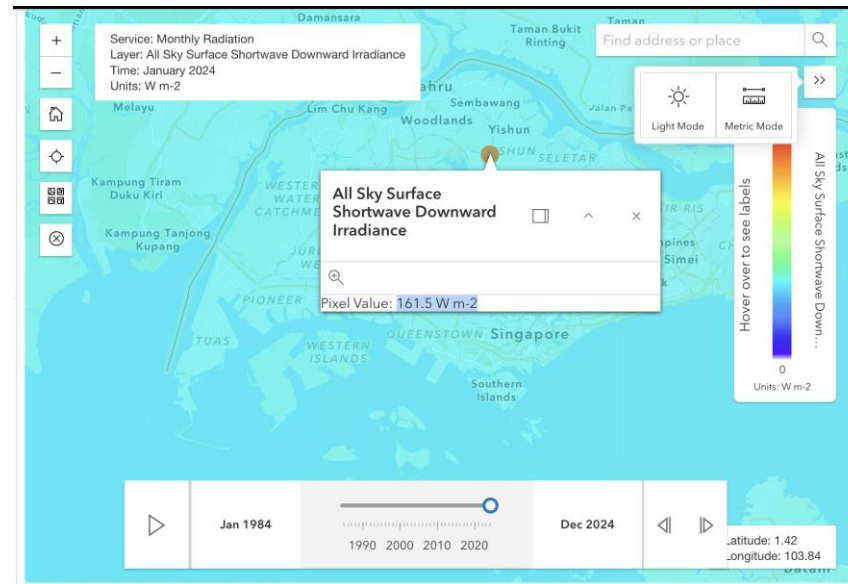
Irradiance (W/m ²)	Current across the Solar Panel (mA)	Voltage across the Solar Panel (V)	Current across the Solar Charge Controller (A)	Voltage across the Solar Charge Controller (V)
400	22.59 	4.915 	4.591 	5.042 
600	34.94 	4.9 	4.750 	4.862 
800	45.45 	4.946 	4.862 	4.750 
1000	55.99 	4.915 	5.042 	4.591 

Figure 4: Peak Sun Hour used to calculate Power Consumption



	January	February	March	April	May	June	July	August	September	October	November	December
Peak Sun Hour	3.876	5.112	5.424	4.788	4.38	4.188	4.656	4.332	4.608	4.488	3.96	4.28

Figure 5: Battery and Solar Panel Sizing Calculations

Power Consumption Device Voltage: 2.553V (Recorded), 3V (Theoretical) Device Current Draw: 290μA	
<u>Battery Sizing</u> i. Nominal system voltage: 3V ii. Days of autonomy: 5 days iii. Total daily load (A x hour of operation/day): 0.00174 Ah/day iv. Unadjusted battery capacity (Days of autonomy x total daily load): 0.0087 Ah v. Maximum allowable depth of discharge (MDOD): 20% vi. Capacity adjusted for MDOD (iv $\div$ v): 0.0435 Ah vii. Maximum daily depth of discharge (MDDOD): 10% viii. Capacity adjusted for MDDOD (iii $\div$ vii): 0.0174 Ah ix. Percent of capacity at end of life (EOL): 80% x. Capacity adjusted for EOL (iv $\div$ ix): 0.010875 Ah xi. Capacity adjusted for depths of discharge or end of life (greatest of vi, viii, or x): 0.0435 Ah xii. Design margin factor (≥ 1): 1.1 xiii. Capacity adjusted for design margin (xi $\times$ xii): 0.04785 Ah xiv. Determine the smallest practical capacity to meet the capacity adjusted for design margin value calculated in xiii: 65 Ah	<u>Solar Panel Sizing</u> Record the voltage required for the WBGT: 3 V Record the current required for the WBGT: 0.00029A 3. Nominal system voltage: 3V 4. Days of autonomy desired: days: 5 5. Total daily load (may be obtained from Part iii - Battery Sizing): 0.00174 Ah/Day 6. Battery capacity (may be obtained from Part xiv - Battery Sizing): 0.065 Ah 7. Assumed total system losses 30% 8. Determine the peak sun hours: 3.876 h 9. Decide on A:L: 1.3 10. Choose a PV Module: Alilin Solar Panel 6V a) Current at Maximum Power (I _{amp}): 0.1 A b) Short Circuit Current (I _{sc}): 0.11 A c) Open Circuit Voltage (V _{oc}): 6.2 V d) Voltage at Maximum Power (V _{mp}): 6 V e) Maximum Power (P _{max}): 0.6 W 11. Daily PV Ah needed (line 5 * 9) : 0.002262 Ah 12. Line 7 / 100, subtracted from 1: 0.7 13. Daily PV Ah per chosen solar panel: 0.27132 Ah 14. Number of panels required: 0.008337019 15. Round up no. of panels required: 1

Figure 6: Power Consumption Measurement

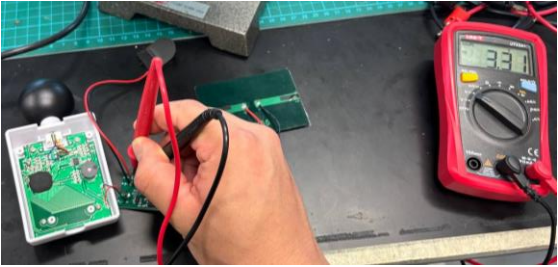
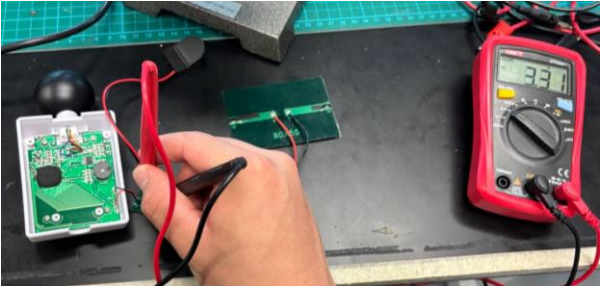
Voltage across Battery (Power Supply to Device by Battery)	Voltage across Load (Power Drawn by Load)	Voltage across Battery (Current Output by Battery)
3.31 V 	3.31 V 	0.15 mA 

Figure 7: Charging Rate Calculation at Differing Irradiance

Irradiance (W/m ²)	Capacity Charged in 2 hour	> than Total Daily Load (1.74mAh)?	> than MDOD (13mAh)?	Time Taken to Charge up Total Daily Load (6 hrs of usage)
400	$45.91 \times 2 = 91.82 \text{ mAh}$	✓	✓	$1.74 \div 45.91 \approx 0.0379 \text{ h} \approx \mathbf{2.27 \text{ min}}$
600	$47.5 \times 2 = 95 \text{ mAh}$	✓	✓	$1.74 \div 47.5 \approx 0.0366 \text{ h} \approx \mathbf{2.20 \text{ min}}$
800	$48.62 \times 2 = 97.24 \text{ mAh}$	✓	✓	$1.74 \div 48.62 \approx 0.0358 \text{ h} \approx \mathbf{2.15 \text{ min}}$
1000	$50.42 \times 2 = 100.84 \text{ mAh}$	✓	✓	$1.74 \div 50.42 \approx 0.0345 \text{ h} \approx \mathbf{2.07 \text{ min}}$

*Reemphasis: These measurements are not reflective of the ideal charging state due to the false information on the technical specification of the only commercially sound solar charge controller. Ideally, the solar charge controller needs to regulate the voltage to 3.2V and provide reasonable charging current of 30mAh.

[Important] With the manufacture of a suited solar charge controller, the time taken to charge up its daily load is $1.74\text{mAh}/30\text{mA} = 0.058\text{h} \approx 3.48 \text{ min}$ ✓, still ideal for the military use case.

Figure 8: Charging Rate Calculation at Differing Irradiance

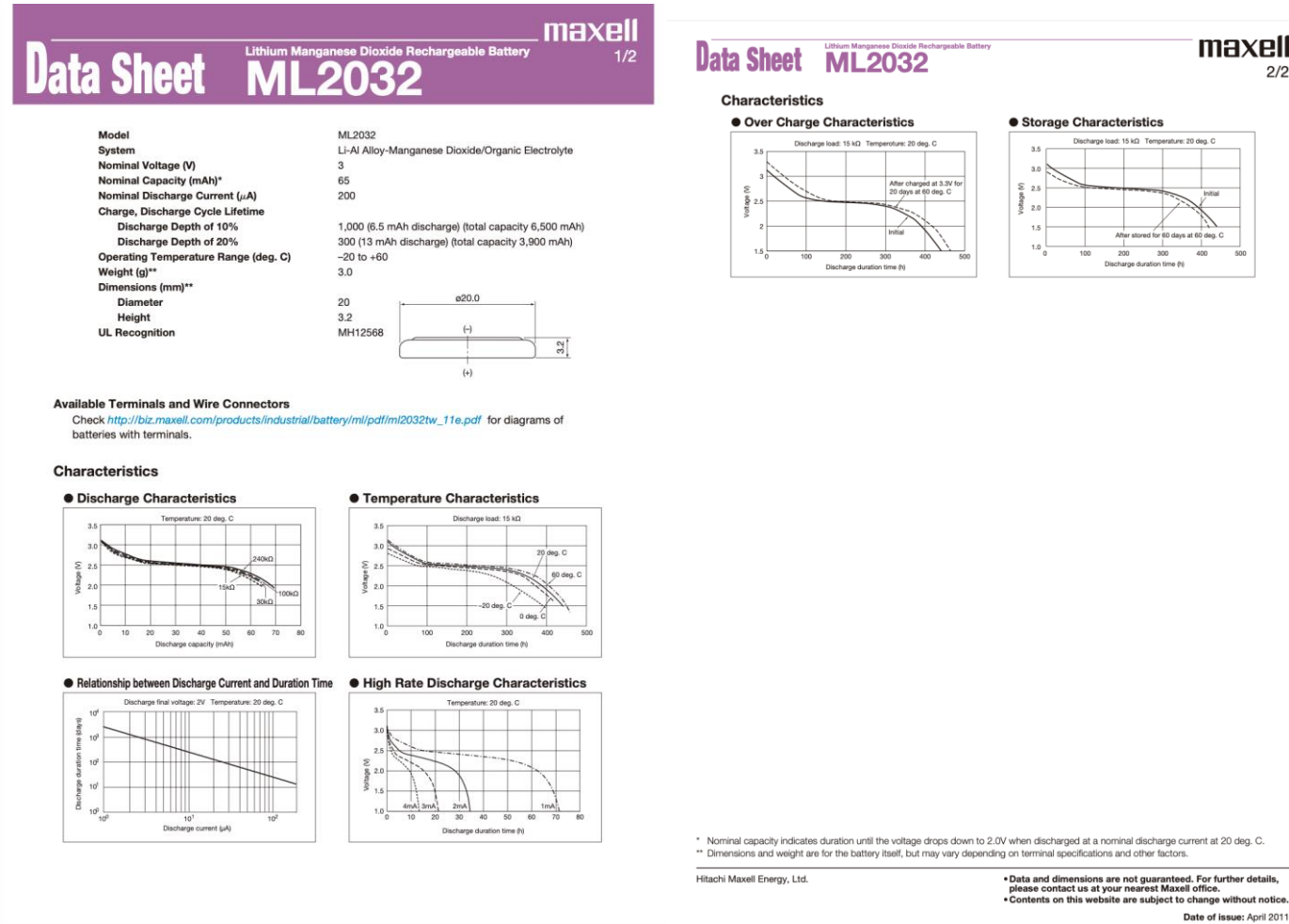
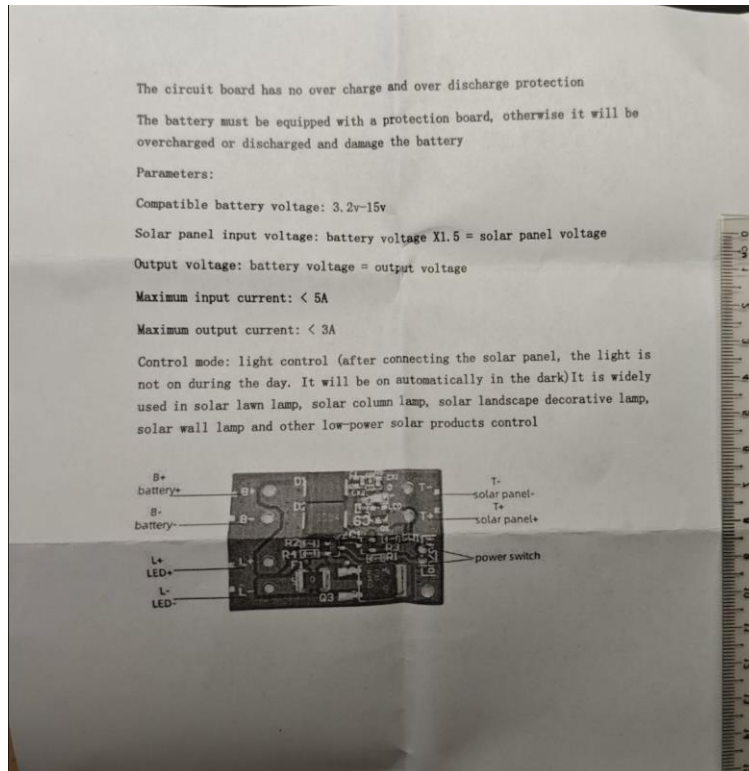


Figure 9: False Specifications on Solar Charge Controller



Product Description

Features:

- 1.Support voltage input between 3.2V-15V, compatible with a variety of solar panels and batteries
2. Selection of high-quality electronic components, product quality assurance
3. Exquisite and compact appearance, components shot beautifully
4. Integrated PCBA board, simple wiring, convenient installation, reduce maintenance costs
5. Low power consumption design, extend the service life of lamps and lanterns, suitable for courtyard and landscape lighting.

Parameter:

Compatible battery voltage: between 3.2V-15V batteries

Input solar panel voltage: X1.5 times the voltage of the connected battery is the voltage of the solar panel to be matched.

Output voltage: the output is as many as the connected V battery.

Input current: <5A

Discharge current: <3A

Control mode: light control control (after connecting the solar panel, the light does not light up during the day, and lights up automatically when it gets dark)

Attention:

1. The output of this board can not be connected to all types of boosting circuits.
2. LED lamp beads should not be used directly, need to be connected to the buck driver (or series with a suitable resistor to limit current)
3. Batteries must be added to protect the board, otherwise it will be overcharged or overdischarged and damaged batteries
4. The board does not have overcharge and overdischarge protection, the battery must have its own protection board, otherwise it will be overcharged or overdischarged and damaged batteries

Package include:

Control Module *1